

T-Shirt Talk: The Geometry of Continued Fractions

Toyesh Jayaswal, Hahn Lheem

PROMYS 2024

The continued fractions thread on the number theory problem sets has always been so elusive. For example, the Super Magic Box is convenient, but where do the periodic and palindromic patterns of the continued fractions for $\sqrt{d}$ come from? Are there better ways to understand the various error bounds for continued fraction convergents?

If you've ever talked with Toyesh about math, you would know that much insight and intuition comes from geometry. This talk will attempt to elucidate some properties of continued fractions by providing an interpretation of continued fraction representations in hyperbolic geometry.

1 Hyperbolic Space

Hyperbolic space is an alternative type of geometry where the set of points is the upper half plane

$$\mathbb{H} = \{(x, y) \in \mathbb{R}^2 \mid y > 0\}$$

and hyperbolic lines are defined as semicircles on the upper half plane which are orthogonal to the x -axis, or vertical lines, which can be interpreted as semicircles passing through the point ∞ . There are many geometric properties of hyperbolic space which make it analogous to Euclidean space in many ways. If you want to learn more about the geometric features of hyperbolic space, take a look at the hyperbolic space talk in the yearbook.

1.1 $\mathrm{SL}_2(\mathbb{R})$

In this section, we discuss the automorphisms of the upper half plane i.e. the transformations of the upper half plane that preserves its geometry. The automorphisms of the Euclidean plane, for example, are translations, dilations, and reflections. In order to find the automorphisms of the upper half plane, we study the transformations of the upper half plane that preserve the hyperbolic lines described above. Oftentimes, the

hyperbolic plane is described via a metric, but it turns out that the metric can be recovered (up to scale factor) by looking at the hyperbolic lines. Thus, if a transformation preserves hyperbolic lines, it is a symmetry.

We have a mysterious action of $\mathrm{SL}_2(\mathbb{R})$ on the upper half plane as follows: we can consider $\mathbb{R}^2$ as $\mathbb{C}$ and thus think of $\mathbb{H}$ as a subspace of $\mathbb{C}$. Now, we can construct the space $\mathbb{CP}^1$ as equivalence classes of pairs of complex numbers $\begin{bmatrix} z \\ w \end{bmatrix}$, not both zero, where two pairs are equivalent if they differ by a nonzero scale factor i.e. $\begin{bmatrix} z \\ w \end{bmatrix} \sim \begin{bmatrix} \lambda z \\ \lambda w \end{bmatrix}$ for $\lambda \in \mathbb{C}^\times$. Similar to $\mathbb{RP}^1$ (refer to the projective geometry notes), $\mathbb{CP}^1$ can be considered as a copy of $\mathbb{C}$ adjoined a point at infinity. Any finite point is equivalent to a point of the form $\begin{bmatrix} z \\ 1 \end{bmatrix}$ and the point at infinity is $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. We will use these identifications going forward treating z and $\begin{bmatrix} z \\ 1 \end{bmatrix}$ as the same thing. A way this is often pictured is the Riemann sphere, where the complex plane is folded up and a point at infinity is added to compactify it into a sphere.

Now, $\mathrm{GL}_2(\mathbb{C})$ acts on $\mathbb{CP}^1$ as follows:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} z \\ 1 \end{bmatrix} = \begin{bmatrix} az + b \\ cz + d \end{bmatrix} \sim \begin{bmatrix} \frac{az+b}{cz+d} \\ 1 \end{bmatrix}.$$

We can bypass the matrix notation and just let the matrix $\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ act on complex numbers z by

$$z \mapsto \gamma z = \frac{az + b}{cz + d}.$$

It is nontrivial to show, but these transformations correspond to (compositions of) translations, scalings, and circle inversions in the plane (hyperbolic geometry notes). In particular, these transformations send circles/lines in $\mathbb{R}^2$ to other circles/lines. In general, these do not preserve the upper half plane, but miraculously, if we restrict to the action by $\mathrm{GL}_2(\mathbb{R})$, the upper half plane is preserved. If $a, b, c, d \in \mathbb{R}$,

$$\mathrm{Im} \left(\frac{az + b}{cz + d} \right) = \mathrm{Im} \left(\frac{(az + b)(c\bar{z} + d)}{|cz + d|^2} \right) = \frac{1}{|cz + d|^2} (ad - bc) \mathrm{Im}(z)$$

and so, if $\mathrm{Im}(z) > 0$, $\mathrm{Im}(\gamma z) > 0$ if $\gamma \in \mathrm{GL}_2(\mathbb{R})$. We recognize $ad - bc$ as the determinant of γ . Note that if we scale γ by a constant factor, it does not change γz , and so we can restrict our attention to matrices with determinant one, which make some computations much cleaner. Thus, we consider the action of $\mathrm{SL}_2(\mathbb{R})$ on $\mathbb{H}$ as a set of symmetries of the upper half plane (actually, these are only the orientation preserving automorphisms, but these are the ones that are relevant later). There is still some redundancy since we can still scale our matrices by ± 1 , but unlike the determinant, there is not a nice (continuous) way to choose a representative in each of these classes. Thus, we can consider the group $\mathrm{PSL}_2(\mathbb{R}) = \mathrm{SL}_2(\mathbb{R}) / \{\pm I\}$. Without proof, it turns out that

$$\mathrm{Aut}(\mathbb{H}) = \mathrm{PSL}_2(\mathbb{R}),$$

or in other words, these are all the transformations that preserve the geometry of the upper half plane. For convenience of using explicit matrices, however, it is often convenient to just work with SL_2 and remember that there is a redundancy.

This may seem quite abstract, so we now discuss an alternative construction which nicely exhibits this action. For this, consider the disc model for the hyperbolic plane (hyperbolic geometry notes). The boundary circle of the disc can be considered as a copy of $\mathbb{RP}^1$ which parametrizes lines in $\mathbb{R}^2$. It should be thought of as the quotient of the unit circle in $\mathbb{R}^2$ by identifying antipodal points (every line through the origin in $\mathbb{R}^2$ intersects the unit circle at antipodal points). Now, any matrix in $GL_2(\mathbb{R})$ sends a line through the origin in $\mathbb{R}^2$ to another line through the origin. Thus, it induces a map S^1 to S^1 which sends antipodal points to antipodal points and thus induces a map $\mathbb{RP}^1$ to $\mathbb{RP}^1$. If we scale a matrix by a positive scale factor, it simply scales all the resulting lines, but they still intersect the unit circle at the same point. Thus, if we only look at orientation preserving actions, we can restrict the action to the action of $SL_2(\mathbb{R})$ without losing any information. Note, if we look at the action of $-I$, it swaps the points on the unit circle at which the lines intersect, so the map from $S^1 \rightarrow S^1$ is not the identity, but the induced map $\mathbb{RP}^1 \rightarrow \mathbb{RP}^1$ is still the identity. Thus, we have that $PSL_2(\mathbb{R})$ acts on $\mathbb{RP}^1$.

Now, any hyperbolic line in the disc model is uniquely determined by its end points. Thus, a map $\mathbb{RP}^1 \rightarrow \mathbb{RP}^1$ extends to a map from hyperbolic lines to hyperbolic lines in the disc model. Such a map in fact uniquely determines a map from $\mathbb{H}$ to $\mathbb{H}$. This is a nontrivial fact since even though we know how what hyperbolic lines get sent to, it is not so clear where points get sent to since a map from a line to a line can be stretched/translated. However, it turns out that all orientation preserving symmetries of the upper half plane can be generated by compositions of reflections (circle inversions) through hyperbolic lines. Orientation preserving transformations fall into 3 categories (hyperbolic geometry notes):

- translation along a hyperbolic line
- rotation around a point (these are the only transformations that have a fixed point)
- transformations which in the upper half plane model of hyperbolic space look like horizontal translations (sometimes called parabolic transformations)

That translations are compositions of reflections through hyperbolic lines is explained in the hyperbolic geometry notes. That the others are also compositions of reflections is left as an exercise.

Now, a circle inversion of the Poincare disc restricted to the boundary circle ($\mathbb{RP}^1$) is also a circle inversion in $\mathbb{RP}^1$, where here, rather than a circle in the plane $\mathbb{R}^2$, we invert through an S^0 - a discrete pair of points - in $\mathbb{R}$. It turns out that inversions in $\mathbb{RP}^1$ are induced by PSL_2 matrices. For example, inverting through the "unit circle" (± 1) is just the map $x \rightarrow \frac{1}{x}$, which is induced by the matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. This is orientation reversing, but we can compose with a flip to get an orientation preserving action. Now, through any two points on the boundary of the Poincare disc, there is a unique geodesic, so we extend the action of inverting through two points p, q in $\mathbb{RP}^1$ to inverting through L

which is the unique hyperbolic line connecting the two points, we can extend the action to $\mathbb{H}$. Since any orientation preserving transformation can be generated by reflections, we can extend the action of any matrix in $\mathrm{GL}_2(\mathbb{R})$ from a map $\mathbb{RP}^1 \rightarrow \mathbb{RP}^1$ to all of $\mathbb{H} \rightarrow \mathbb{H}$ by decomposing into reflections. That this is well defined is left as an exercise. Thus, we obtain an action of $\mathrm{PSL}_2(\mathbb{R})$ on $\mathbb{H}$ which is fairly direct in that it does not introduce complex numbers which are not as directly connected to geometry.

2 Farey Tessellation

The $\mathrm{SL}_2(\mathbb{R})$ -action on $\mathbb{H}$ gives us a complete picture of all possible symmetries on the upper half plane. As number theorists, though, we are interested in things discrete, especially $\mathbb{Z}$. So what if we only look at the $\mathrm{SL}_2(\mathbb{Z})$ -action on the upper half plane? What kind of interesting symmetries can we see?

Let's do an example in $\mathbb{R}^2$ for motivation. The symmetries of the plane consist of translations and rotations: we can translate by any vector in $\mathbb{R}^2$ or rotate by any angle $\theta \in 2\pi\mathbb{R}$, and this will preserve all distances in the plane.

Suppose, instead of allowing for translation by all of $\mathbb{R}^2$ or rotation by any angle in $\mathbb{R}$, we focused on the symmetries related to $\mathbb{Z}$. For instance, suppose we allow only left and right translations by an integer, and we restrict rotations to only rotation by $\pi/4$ radians counterclockwise with respect to the origin. Take the line $x = 0$ and consider its image under all of these symmetries. What do we get? I claim we can get all of the lines $x = n$ and $y = m$, where $n, m \in \mathbb{Z}$.

Exercise 2.1. Start with $x = 0$. Using just left and right translations by an integer and $\pi/4$ -radian rotations around the origin, can you get to the line $y = -17$? Can you show that no other lines can be reached other than $x = n$ or $y = m$ for integers $n, m \in \mathbb{Z}$?

Taking all the images of $x = 0$ under these symmetries in consideration, we have provided a tiling of the plane $\mathbb{R}^2$ into unit squares! The corners of the tiles give us a lattice in $\mathbb{R}^2$, which we can easily see to be the integral points $\mathbb{Z}^2$. Furthermore, we can get from any one unit square to another through these rotations and translations.

Even though the upper half plane has a vastly different geometry (straight lines in $\mathbb{H}$ are what we typically consider as semicircles), we can hope to achieve the same thing. We have established $\mathrm{SL}_2(\mathbb{R})$ as all the isometries (distance-preserving automorphisms) of $\mathbb{H}$. We wish to replicate the same phenomenon in $\mathbb{R}^2$: take some line, look at its images under the discrete $\mathrm{SL}_2(\mathbb{Z})$ -action, and obtain a tiling of $\mathbb{H}$ with specified "lattice" points of interest.

We will start with the straight vertical line from 0 to ∞ . Denote it $\mathbb{I}$, as this is the imaginary axis in $\mathbb{H}$. It is one of the geodesics described in the previous section.

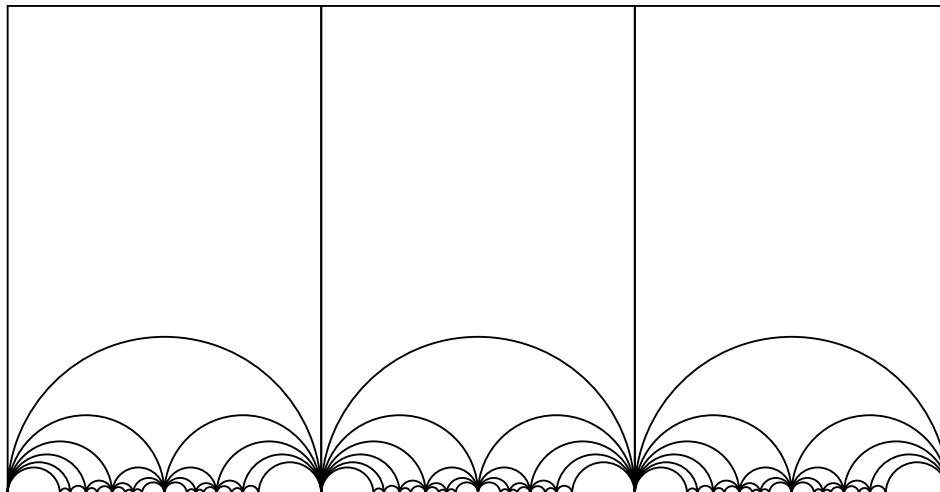
The $\mathrm{SL}_2(\mathbb{R})$ -action maps lines to lines, so we are interested in where some $\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ sends our line $\mathbb{I}$ from 0 to ∞ . Any line is determined by two points, so

it suffices to see the images of the endpoints 0 and ∞ , and this will uniquely determine the line. By definition of our action, we have

$$\begin{aligned}\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot 0 &= \frac{a \cdot 0 + b}{c \cdot 0 + d} = \frac{b}{d} \\ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \infty &= \frac{a \cdot \infty + b}{c \cdot \infty + d} = \frac{a}{c}.\end{aligned}$$

(I realize I would be crucified if I wrote the second equation on any PROMYS problem set, but bear with us here. Students who did the “Fractional Linear Functions” exploration lab will understand.)

The computation above tells us that any image of $\mathbb{I}$ under some $\mathrm{SL}_2(\mathbb{Z})$ -action will be the hyperbolic line (i.e., semicircle or vertical line) connecting the fractions $\frac{a}{b}$ and $\frac{c}{d}$, where $ad - bc = 1$ since $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Note if $b = 0$ (respectively, $d = 0$), then $\frac{a}{b} = \infty$ (resp., $\frac{c}{d} = 0$), meaning the line is vertical. Furthermore, if $b = 0$ and $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z})$, we must have $a = d = \pm 1$, so the line goes from some $\pm c \in \mathbb{Z}$ to ∞ . Below is a picture of all these images:



If we imagine all vertical lines intersecting each other at ∞ , we can see that every “fundamental domain” (i.e., every region which cannot be split into smaller ones) is a *triangle* in hyperbolic space, as it is bounded by three straight lines. The $\mathrm{SL}_2(\mathbb{Z})$ -action on $\mathbb{H}$ therefore gives us a tessellation of $\mathbb{H}$ into triangles.

We chose to restrict ourselves to $\mathrm{SL}_2(\mathbb{Z})$ to see some number theory. Where is the number theory here? Consider all endpoints on the real axis. We already know from the action that $\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ maps 0 to $\frac{c}{d}$ and ∞ to $\frac{a}{b}$, so the set of endpoints of these geodesics on the real line is exactly the rationals $\mathbb{Q}$. This is pretty neat: just by looking at the images of one vertical line in $\mathbb{H}$, we got all rational numbers.

Seeing which rationals are connected to each other by some geodesic reveals even more information in this tessellation. The endpoints of any geodesic are $\{\frac{a}{b}, \frac{c}{d}\}$ with

$ad - bc = 1$, or equivalently $\left| \frac{a}{b} - \frac{c}{d} \right| = \frac{1}{bd}$. From the earlier number theory problem sets, you may recall that this property of differences between consecutive fractions is satisfied by the Farey sequences! Therefore, these geodesics are actually connecting consecutive fractions in some Farey sequence.

Example 2.2. Consider the Farey sequence F_7 consisting of fractions with denominator at most 7. One can verify that $\frac{2}{5}$ and $\frac{3}{7}$ are consecutive terms in F_7 . Indeed, the matrix $\begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ sends the line $\mathbb{I}$ to the semicircle geodesic connecting $\frac{2}{5}$ with $\frac{3}{7}$, with $0 \mapsto \frac{2}{5}$ and $\infty \mapsto \frac{3}{7}$.

Within the Farey sequences are continued fraction convergents. Let $\alpha \in \mathbb{R}$, and let P_k/Q_k and P_{k+1}/Q_{k+1} are consecutive continued fraction convergents of α . Problem #1 from Problem Set 5 tells us that the determinant

$$\begin{vmatrix} P_k & P_{k+1} \\ Q_k & Q_{k+1} \end{vmatrix} = (-1)^k,$$

so $\frac{P_k}{Q_k}$ and $\frac{P_{k+1}}{Q_{k+1}}$ are consecutive terms in the Q_{k+1} -Farey sequence. In other words, they are connected by some geodesic in the Farey tessellation.

The hope, therefore, is to be able to read off the continued fraction convergents for some $\alpha \in \mathbb{R}$ through this geometric picture.

Remark 2.3 (A long aside on modular curves). If slapping on $\mathbb{Z}$ in $\mathrm{SL}_2(\mathbb{Z})$ and the appearance of the Farey sequences is not enough for your number theory vices, here is another significant connection to number theory. Given a space like $\mathbb{H}$ with a group action like $\mathrm{SL}_2(\mathbb{Z})$ via linear fractional transformations, one can partition $\mathbb{H}$ into disjoint regions where for each region, given any $z \in \mathbb{H}$, there exists a unique $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that $\gamma \cdot z$ is in the region. If we were to do the same in Euclidean space $\mathbb{R}^2$ with a $\mathbb{Z} \oplus \mathbb{Z}$ -action given by $(m \oplus n) \cdot (x, y) = (x + m, y + n)$, then the disjoint regions would just be the unit squares in $\mathbb{R}^2$, i.e., the fundamental domain of the lattice $\mathbb{Z} \oplus \mathbb{Z}$.

Denote one of the disjoint regions obtained from the $\mathrm{SL}_2(\mathbb{Z})$ -action on $\mathbb{H}$ as the quotient $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$. (For those who attended Carol and Henry's group theory minicourses, this is the same idea as cosets and quotient groups in group theory, except more generally we can take cosets on sets with a group action.) This region may seem awfully insignificant and awkward at first, but its importance cannot be overstated. This is not really the place to go into details, but among other reasons, this quotient space is notable because it somehow *parametrizes all complex elliptic curves* (peep Angeline's minicourse on elliptic curves!) via a very important complex analytic function called the j -invariant. Functions on this space can be seen as *modular forms*, which was the topic of one of the 2023 advanced seminars, taught by the Overlord herself Li-Mei Lim.

We can replace $\mathrm{SL}_2(\mathbb{Z})$ with a subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$, and as long as Γ is reasonably chosen, the quotient space $\Gamma \backslash \mathbb{H}$ (where the action is just restricting the $\mathrm{SL}_2(\mathbb{Z})$ -action to only Γ) also has relations to elliptic curves and modular forms. Even better, all these

spaces have the additional structure of a *Riemann surface*, a nice complex manifold whose theory is lovely and enriching.

I go on this long tangent because each ideal triangle in the Farey tessellation is exactly the quotient space $\Gamma_0(2)\backslash\mathbb{H}$, where $\Gamma_0(2) \subset \mathrm{SL}_2(\mathbb{Z})$ is the subgroup consisting of matrices

$$\Gamma_0(2) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : c \equiv 0 \pmod{2} \right\}.$$

So although we are describing a really beautiful and nifty connection between these Farey sequences and continued fractions in this talk/handout, the Farey tessellation contains an impressive amount of arithmetic significance that shouldn't be understated.

3 From Cutting Sequence to Continued Fraction

We will determine the continued fraction sequence for some $\alpha \in \mathbb{R}$ geometrically. We will keep track of a *cutting sequence*, which will be just a sequence of R's and L's, and this will determine the continued fraction sequence.

1. On the Farey tessellation, draw a geodesic from a point on the imaginary axis $\mathbb{I}$ to $\sqrt{6}$. It does not matter where on $\mathbb{I}$ you start. An example geodesic is drawn below (as well as on the T-shirt) for $\alpha = \sqrt{6}$, where the leftmost vertical line is $\mathbb{I}$.
2. Traverse along the geodesic from $\mathbb{I}$ towards α . Every time the geodesic crosses one of the ideal triangles in the Farey tessellation, it intersects two of its sides.
3. If these two sides of the triangle intersect to the left of your geodesic, add an "L" to the cutting sequence. If they intersect to the right, add an "R" to the cutting sequence.

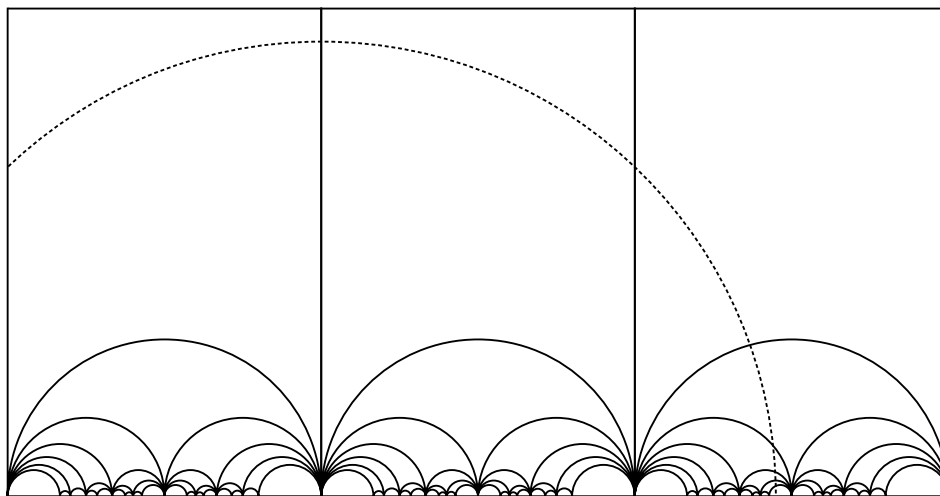


Figure 1: Geodesic for $\sqrt{6}$ crossing through the Farey Tessellation.

We will demonstrate the first few terms in our cutting sequence for $\alpha = \sqrt{6}$. The first ideal triangle crossed is the one in the leftmost third of the figure extending towards ∞ . Since the two vertical sides intersect at ∞ , the intersection point is to the *left* of the geodesic. (Imagine yourself traveling along the geodesic from left to right.) Thus, the first letter in the cutting sequence is **L**. Similarly, the second ideal triangle crossed also extends to ∞ , so the vertical sides intersect to the *left* of the geodesic. We append an L to the cutting sequence, making it **LL**.

The third ideal triangle crossed is the one extending to ∞ in the rightmost third of the figure, except now the geodesic intersects the left and bottom sides. These two sides intersect at $z = 2$, which is to the *right* of the geodesic. Now our cutting sequence looks like **LLR**.

Exercise 3.1. Verify from the figure that the first seven letters in the cutting sequence for $\sqrt{6}$ are LLRRLLL.

Although our figure is too small to draw the ideal triangles closer to the real axis, we could continue the process and obtain **LLRRLLLLRRLLLL...** as the first few letters of the cutting sequence for $\sqrt{6}$. (This is written on the front of the T-shirt.) Now, it is incredibly easy to read off the continued fraction representation for $\sqrt{6}$: count the number of consecutive instances of the same letter. In detail,

1. The sequence begins with two L's, so $a_1 = 2$.
2. The next sequence of repeating letters is RR, so $a_2 = 2$.
3. The next string of LLLL mandates $a_3 = 4$.
4. In the same vein, RR gives $a_4 = 2$ and LLLL gives $a_5 = 4$.

Exercise 3.2. Verify that, indeed as written on the T-shirt,

$$\boxed{\sqrt{6} = [2; \overline{2, 4}]}$$

Perhaps this seems magical to you. But like with the magic box, the magic becomes intuition if one thinks deeply enough. So we will think deeper on the relationship between these Farey geodesics and continued fractions.

The first is that the mechanism running behind both the Farey geodesics and the magic box is matrix multiplication. To refresh on matrix multiplication, we have

$$\begin{aligned} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix} \\ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} x & y \\ z & w \end{bmatrix} &= \begin{bmatrix} ax + bz & ay + bw \\ cx + dz & cy + dw \end{bmatrix}. \end{aligned}$$

First, we can reformulate our computations for finding the image of a point under some $\text{SL}_2(\mathbb{Z})$ -action via matrices. Previously, we defined the group action as $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot z =$

$\frac{az+b}{cz+d}$. But viewing $\mathbb{H} \subset \mathbb{P}^1(\mathbb{C})$ and the endpoints of the Farey geodesics as points in $\mathbb{P}^1(\mathbb{Q}) = \mathbb{Q} \cup \{\infty\}$, we see that this group action is just matrix multiplication:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} z \\ 1 \end{bmatrix} = \begin{bmatrix} az + b \\ cz + d \end{bmatrix}.$$

Second, let P_k/Q_k denote the k^{th} continued fraction convergent for some $\alpha \in \mathbb{R}$. Problem #1 of Problem Set 5 tells you that there is a nice recurrence relation between consecutive convergents. Explicitly, if $\alpha = [a_1, a_2, \dots]$, then

$$P_{k+1} = a_k P_k + P_{k-1}, \quad Q_{k+1} = a_k Q_k + Q_{k-1}.$$

This begs to be seen as matrix multiplication. Start with the identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ – you can view the columns as the 0^{th} and $(-1)^{\text{th}}$ continued fraction convergents. Then, multiplying by $\begin{bmatrix} a_1 & 1 \\ 1 & 0 \end{bmatrix}$ returns us the same matrix, and we can see the columns as the first and zeroth convergents. Doing one more to paint the picture, we have

$$\begin{bmatrix} a_1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a_2 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} a_1 a_2 + 1 & a_1 \\ a_2 & 1 \end{bmatrix} = \begin{bmatrix} P_2 & P_1 \\ Q_2 & Q_1 \end{bmatrix}.$$

Exercise 3.3. Can you generalize? Can you prove? With this perspective, prove $P_{k+1}Q_k - Q_{k+1}P_k = (-1)^k$.

We seek for a greater understanding of the $\begin{bmatrix} a_i & 1 \\ 1 & 0 \end{bmatrix}$ matrices. Just before, we began to see endpoints of Farey geodesics as living in $\mathbb{P}^1$, hence representable as column vectors. So it is natural to interpret a square matrix as holding the data of two points. Indeed, we can interpret any $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \text{SL}_2(\mathbb{Z})$ as the geodesic connecting $\frac{a}{b}$ and $\frac{c}{d}$. Matrix multiplication can now be viewed as sending one geodesic in the Farey tessellation, represented as an element of $\text{SL}_2(\mathbb{Z})$, to another geodesic. The magic box, as seen through matrix multiplication, is now describing how paths between consecutive continued fraction convergents are sent to one another. This movement between consecutive geodesics is encapsulated in these $\begin{bmatrix} a_i & 1 \\ 1 & 0 \end{bmatrix}$ matrices.

We will write such matrices instead as

$$\begin{bmatrix} a_i & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & a_i \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

This seems like an inconsequential step, but this makes the group action far clearer. Observe that

$$\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} x + a \\ 1 \end{bmatrix},$$

so the first matrix in the product on the right is simply translation by a_i . The $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ matrix, in contrast, is the inversion map sending $x \mapsto 1/x$. Although this is not

exactly suitable for our purposes as $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \notin \mathrm{SL}_2(\mathbb{Z})$, it motivates us to see this geometric procedure through translations and inversions.

We return to the $\alpha = \sqrt{6}$ example. The first $\mathbf{L}$ comes from intersecting another vertical line. This is simply a translation of the imaginary axis $\mathbb{I}$, so we have in effect acted on the Farey tessellation by $T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. The next $\mathbf{L}$ is the same, so the first two $\mathbf{L}$'s amounts to acting on the tessellation by

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

The next intersection with the tessellation is no longer with a vertical line; it is no longer clear how to get from the left side of the ideal triangle with the bottom side, as it is not just translation by 1. To return back to Kansas, we apply the inversion $S = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ on the entire Farey tessellation, including our geodesic. As $S \in \mathrm{SL}_2(\mathbb{Z})$ and the action is on all geodesics, all orientations are preserved and all intersections between geodesics remain as intersections.

The benefit of this action is that now, non-vertical lines return as vertical lines. For instance, the geodesic originally connecting 2 and 3, which now connects 0 and 1 after the two translations, gets mapped to

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix},$$

which corresponds to the vertical line between -1 and ∞ . In fact, all geodesics with endpoint originally at $z = 2$ are now vertical lines under S .

The catch is that S also changes direction, so now instead of traveling to the right towards $\sqrt{6}$, we are traveling to the left towards $S \cdot (\sqrt{6} - 2) = -\frac{1}{\sqrt{6}-2}$, which closely mimics the computations one would make if they were to compute the continued fraction representation for $\sqrt{6}$ manually. Because the direction of the geodesic is reversed but all orientations are preserved (e.g., what was to the right before remains to the right), the intersections at ∞ now correspond to R's in the cutting sequence. We can continue this ad infinitum.

The number of times we can translate, e.g., the number of consecutive L's or R's, is dictated by the floor of the approximating irrational. For instance, we were allowed two translations because $\lfloor \sqrt{6} \rfloor = 2$, and likewise the next consecutive sequence has two R's because $\left\lfloor \frac{1}{\sqrt{6}-2} \right\rfloor = 2$.

Exercise 3.4. Suppose your geodesic to α intersects an ideal triangle at two sides, and let p/q be the point of intersection of the two sides. Show that p/q is a continued fraction convergent for α .

Finally, we have a clear path between continued fraction convergents. Note, as suggested before, that the matrices $\begin{bmatrix} a_k & 1 \\ 0 & 1 \end{bmatrix}$ satisfy

$$\begin{bmatrix} a_k & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} P_k & P_{k-1} \\ Q_k & Q_{k-1} \end{bmatrix} = \begin{bmatrix} P_{k+1} & P_k \\ Q_{k+1} & Q_k \end{bmatrix}.$$

Seeing the matrices of convergents as paths, the multiplication is sending the geodesic between P_{k-1}/Q_{k-1} and P_k/Q_k to the one between P_k/Q_k and P_{k+1}/Q_{k+1} . These are clearly connected at P_k/Q_k , so we have a connected path between all continued fraction convergents which converges to α .

4 NT Problem Sets, revisited

We have touched on several properties of continued fractions featured on the number theory problem sets already, but we can prove them directly using this geometric approach.

Problem 4.1. Let P_k/Q_k be the k^{th} continued fraction convergent of $\alpha \in \mathbb{R}$. Prove that $\begin{vmatrix} P_k & P_{k-1} \\ Q_k & Q_{k-1} \end{vmatrix} = (-1)^k$.

Proof. The procedure tells us that consecutive continued fraction convergents are end-points of some geodesic, so they form a matrix in $\text{SL}_2(\mathbb{Z})$. The sign depends on the order of the columns. $\square$

Problem 4.2. Prove that $P_k/Q_k < \alpha < P_{k+1}/Q_{k+1}$ for odd k and the reverse for even k .

Proof. For $\alpha > 1$, the first convergent corresponds to the first intersection point to the right. Intersection of sides of a traversed ideal triangle correspond to the continued fraction convergents, the intersection being to the left of the geodesic indicates overestimation, and the intersection being to the right indicates underestimation. $\square$

Problem 4.3. If $\left| \alpha - \frac{P_k}{Q_k} \right| > \left| \alpha - \frac{p}{q} \right|$, then $q > Q_k$.

Proof. Because there exists a geodesic connecting P_{k-1}/Q_{k-1} and P_k/Q_k (which we know exists because the geodesic to α must cross it), they must be consecutive terms in the Q_k -Farey sequence. This forces $q > Q_k$. $\square$

Problem 4.4. Prove that

$$\left| \alpha - \frac{P_k}{Q_k} \right| < \frac{1}{Q_k Q_{k+1}}.$$

Proof. This follows immediately from the above problem and the fact that P_k/Q_k and P_{k+1}/Q_{k+1} are consecutive terms in the Q_{k+1} -Farey sequence. $\square$

The above bound is enough to show the existence of a solution to Pell's equation $x^2 - dy^2 = 1$ *without any other facts about continued fractions*. Although I won't spell out the steps due to spoilers, I encourage the reader to visit the following problems: Problem Set #19, Problem 13; Problem Set #21, Problem 12; Problem Set #24, Problems 6,

7, 8. The existence of such a solution will be necessary for our work with continued fractions for $\sqrt{d}$.

Additionally, we will formalize the notion that all isometries of $\mathbb{H}$ come from translations and inversions. Recall that translations by $\mathbb{Z}$ are generated by the matrix $T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, while inversion is given by $S = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

Proposition 4.5. $\text{SL}_2(\mathbb{Z})$ is generated by S and T as a group, i.e., any matrix in $\text{SL}_2(\mathbb{Z})$ can be expressed as a product of S 's and T 's.

Proof. Let $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \text{SL}_2(\mathbb{Z})$. By computing the continued fraction sequence for $\frac{a}{c}$, we can find a matrix $\gamma \in \text{SL}_2(\mathbb{Z})$ such that $\gamma \cdot \begin{bmatrix} a \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Furthermore, γ is expressible as a product of S 's and T 's, where the number of T 's and the placement of S is dictated by the continued fraction sequence of $\frac{a}{c}$. But γ should map $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ to another matrix in $\text{SL}_2(\mathbb{Z})$, so we have forced

$$\gamma \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

and the conclusion follows. $\square$

We find this a particularly slick proof because oftentimes, textbooks will neglect to prove this fact or, if they do, they choose to give a very computationally heavy approach. This required very little work, just a strong understanding of the Farey tessellation and its properties.

5 Periodicity of $\sqrt{d}$

We will now use the developed machinery to prove that for n squarefree, $\sqrt{n}$ is periodic after the leading term, and has an almost palindromic continued fraction. Given a solution to Pell's equation $x^2 - ny^2 = 1$, we can construct an element of $\text{SL}_2(\mathbb{Z})$ which fixes the hyperbolic line L passing through $\pm\sqrt{n}$. Explicitly, we consider

$$\begin{bmatrix} x & ny \\ y & x \end{bmatrix} \begin{bmatrix} \pm\sqrt{n} \\ 1 \end{bmatrix} = \begin{bmatrix} \pm\sqrt{n}x + ny \\ x + (\pm\sqrt{n})y \end{bmatrix} \sim \begin{bmatrix} \pm\sqrt{n} \\ 1 \end{bmatrix}$$

and thus $M = \begin{bmatrix} x & ny \\ y & x \end{bmatrix}$ preserves the endpoints of L ($\pm\sqrt{n}$) and thus all of L . Geometrically, it acts as a translation along this line. Note that we never used that (x, y) was a solution to Pell's equation in the above computation, but it grants us that this matrix is in fact an element of $\text{SL}_2(\mathbb{Z})$.

Looking at its action at the point i :

$$\begin{bmatrix} x & ny \\ y & x \end{bmatrix} \begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} ix + ny \\ x + iy \end{bmatrix} \sim \begin{bmatrix} \frac{xy(n-1) + i(x^2 - ny^2)}{x^2 + y^2} \\ 1 \end{bmatrix}$$

which has positive real part. Thus, we can see that M translates towards the right.

Now we will show that $\lfloor \sqrt{n} \rfloor + \sqrt{n}$ has periodic continued fraction. Notice that the geodesic connecting this number to its conjugate $\lfloor \sqrt{n} \rfloor - \sqrt{n}$ is the hyperbolic line $L' = T^{\lfloor \sqrt{n} \rfloor} L$. Now, the conjugated matrix $M' = T^{\lfloor \sqrt{n} \rfloor} M T^{-\lfloor \sqrt{n} \rfloor}$ fixes L' as it fixes its endpoints, and so is a translation towards the right along this line.

Now, since $M' \in \text{SL}_2(\mathbb{Z})$, it can be written as a product of T 's and S 's. I claim that the form of the matrix is $M' = T^{a_k} \dots T^{a_1} S$ i.e. it starts with a T and ends with an S . We can see this by realizing that $(M')^{-1}$ does some steps of the continued fraction algorithm on $\lfloor \sqrt{n} \rfloor + \sqrt{n}$ and returns L' to itself at the end. Thus, the first transformation of $(M')^{-1}$ must be T^{-1} as $\lfloor \sqrt{n} \rfloor + \sqrt{n} > 1$.

$(M')^{-1}$ must end in an S because of the following: $\lfloor \sqrt{n} \rfloor + \sqrt{n} > 0$ so that the last step would have to be T^{-1} or S . However $-1 < \lfloor \sqrt{n} \rfloor - \sqrt{n} < 0$ so if it were T^{-1} , both ends of the geodesic line must have had to lie on the right side of the imaginary axis in the last step. This is not possible as at each step, the imaginary axis is the image of a line in the Farey sequence that intersects L' . Note that the inequality $-1 < \lfloor \sqrt{n} \rfloor - \sqrt{n} < 0$ is the reason that we're analyzing $\lfloor \sqrt{n} \rfloor + \sqrt{n}$ instead of just $\sqrt{n}$.

Now, for any point $p \in L'$, for large enough N , $(M')^N$ will take p arbitrarily close (in the Euclidean sense) to $\lfloor \sqrt{n} \rfloor + \sqrt{n}$, as it translates to the right. In particular, L' intersects the imaginary axis and the line connecting 1 to ∞ . Thus, as $M' \in \text{SL}_2(\mathbb{Z})$, it sends these two lines to two other lines of a triangle in the Farey sequence which get arbitrarily close to $\lfloor \sqrt{n} \rfloor + \sqrt{n}$. If two sides of a triangle in the sequence are small, the third side must also be small, so for large N , $(M')^N$ takes the fundamental triangle connecting 0, 1, ∞ arbitrarily close to $\lfloor \sqrt{n} \rfloor + \sqrt{n}$. Thus, expanding $(M')^N$ into S 's and T 's must give convergents of this number.

Now,

$$(M')^N = T^{a_k} \dots T^{a_1} S T^{a_k} \dots T^{a_1} S T^{a_k} \dots \dots S T^{a_1} S$$

and since M' starts with a T and ends in and S , the powers of T and S from different factors of M' do not interact. Thus, we see that

$$\lfloor \sqrt{n} \rfloor + \sqrt{n} = [a_1; \overline{a_2, \dots, a_k, a_1}].$$

Thus, we have proven that this number is purely periodic!

Example 5.1. For $\sqrt{11} + 3$, a solution to Pell's equation is $10^2 - 11 * 3^2 = 1$. Thus, we get the matrix $M = \begin{bmatrix} 10 & 33 \\ 3 & 10 \end{bmatrix}$. Analyzing the continued fraction of $33/10$, we find that $M = T^3 S T^{-3} S T^3$. Now, we need to conjugate M by T^{-3} in order to get M' which fixes $\pm\sqrt{11} + 3$. This gives $M' = T^6 S T^{-3} S$. Thus, looking at $(M')^N$, we see that the continued fraction of $\sqrt{11} + 3$ is $[6; \overline{3, 6}]$.

6 Palindromicity of $\sqrt{d}$

We can actually also use these geometric methods to prove that $\sqrt{n}$ is almost palindromic. First, we claim that there is a "minimal" unit $x + y\sqrt{n}$ in $\mathbb{Z}[\sqrt{d}]$, which in our case means the matrix $M = \begin{bmatrix} x & ny \\ y & x \end{bmatrix}$ translates L (the line connecting $\pm\sqrt{n}$) by the smallest amount. The reason for this is the group of units acts as a discrete group of translations on the line L . We can identify L with $\mathbb{R}$ by choosing an arbitrary point p on the line to be 0 and identifying a point $q \in L$ a (signed) hyperbolic distance of d away from p with d . Then, the orbit of the group of units (which is in bijection with the group of units) forms a subgroup of $\mathbb{R}$. Any subgroup of $\mathbb{R}$ is either cyclic or dense, and this one is discrete so it is cyclic (exercises if you don't believe it).

Thus, we can choose a minimal unit $x + y\sqrt{n}$ corresponding to a minimal right-translating matrix M , which we use from now on. The previous section implies that

$$\sqrt{n} = [a_0; \overline{a_1, \dots, a_k}]$$

(the a_i here are different from the last section).

Note that the first few steps of the continued fraction algorithm for $\sqrt{n}$ is given by M^{-1} , or more precisely, its decomposition into translations and inversions. However, by symmetry, we see that applying M would apply the reflection of this continued fraction algorithm for $-\sqrt{n}$. If $\sqrt{n}$ were *purely* periodic, $\sqrt{n} = [\overline{b_1, \dots, b_n}]$, we would be able to say that $M = T^{b_1} S \dots T^{b_n} S$ and thus

$$S(-\sqrt{n}) = \frac{1}{\sqrt{n}} = T^{b_n} S \dots T^{b_1} S \dots = [\overline{b_n, \dots, b_1}].$$

Unfortunately, $\sqrt{n}$ does not have a purely periodic expansion. But we can modify it a little bit so that it does!

$$\sqrt{n} = [a_0; \overline{a_1, \dots, a_k}] = [\overline{a_0, a_1, \dots, a_{k-1}, a_k - a_0, 0}]$$

The reason is that we can write

$$a_k = a_k - a_0 + \frac{1}{0 + \frac{1}{a_0}}$$

and in terms of the matrix M , this corresponds to replacing T^{a_k} with $T^{a_k - a_0} S T^0 S T^{a_0}$. Thus, we have

$$\frac{1}{\sqrt{n}} = [0, \overline{a_k - a_0, a_{k-1}, \dots, a_1, a_0}] = [0; \overline{a_k - a_0, a_{k-1}, \dots, a_1, a_0, 0}] = 0 + \frac{1}{a_k - a_0 + \frac{1}{1 + \dots}}.$$

Thus, we see also that

$$\sqrt{n} = [\overline{a_k - a_0, a_{k-1}, \dots, a_1, a_0, 0}] = [a_k - a_0; \overline{a_{k-1}, \dots, a_1, a_k}]$$

where the last equality is the reverse of the 0 trick we used earlier. By the uniqueness of continued fractions, we see that:

- $a_k - a_0 = a_0$, or $a_k = 2a_0$
- $a_i = a_{k-i}$ for $1 \leq i \leq k-1$

i.e. the continued fraction is almost palindromic! Note that we could see $a_k = 2a_0$ from the last section since the a_k is the same as the one in that section which is $a_k = \lfloor \sqrt{n} \rfloor + \sqrt{n} = 2\lfloor \sqrt{n} \rfloor = 2a_0$.

Another nice thing to note is that the period point corresponds to the matrix M which was constructed by using a solution to Pell's equation. Thus, if P/Q is the convergent at the period point, $P^2 - nQ^2 = 1$ and we can compute a solution to Pell's equation. In fact, the logic can be reversed and it can be argued that the first point a convergent P_m/Q_m gives $P_m^2 - nQ_m^2 = 1$, we get such a translation matrix M , and thus must be the minimal unit in $\mathbb{Z}[\sqrt{n}]$. Hence, we have found a way to explicitly compute all units in $\mathbb{Z}[\sqrt{n}]$ instead of just abstractly proving they exist.

Example 6.1. $\sqrt{61} = [7; \overline{1, 4, 3, 1, 2, 2, 1, 3, 4, 1, 14}]$ which has the form expected!